

# Liangliang Zhang

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## EDUCATION

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**Rensselaer Polytechnic Institute (RPI)**

Troy, NY

*Ph.D. in Computer Science, advised by Prof. Yao Ma*

*Aug. 2024 – May 2028 (Expected)*

**Central South University**

Changsha, China

*B.S. in Software Engineering*

*Sep. 2022 – Jun. 2024*

## RESEARCH INTERESTS

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Large Language Models, Agentic AI, Retrieval-Augmented Generation, Knowledge Graphs, Graph Machine Learning, and Data-Centric AI.

## SELECTED CONFERENCE AND JOURNAL PUBLICATIONS

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- [1] **Zhang, Liangliang** and Y. Ma. “Planning Before Retrieval: A Constrained Planner-Executor Framework for Reliable and Efficient KG-Augmented Reasoning”. In: *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining*. 2026, pp. 1308–1310.
- [2] **Zhang, Liangliang**, Z. Jiang, H. Chi, H. Chen, M. Elkoumy, F. Wang, Q. Wu, Z. Zhou, S. Pan, S. Wang, et al. “Diagnosing and Addressing Pitfalls in KG-RAG Datasets: Toward More Reliable Benchmarking”. In: *In Proceedings of the Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS)* (2025).
- [3] **Zhang, Liangliang**, H. Bao, and Y. Ma. “Extending Graph Condensation to Multi-Label Datasets: A Benchmark Study”. In: *Transactions on Machine Learning Research (TMLR)* (2025).
- [4] H. Xu, **Zhang, Liangliang**, Y. Ma, S. Zhou, Z. Zheng, and J. Bu. “Learning to Reduce the Scale of Large Graphs: A Comprehensive Survey”. In: *ACM Transactions on Knowledge Discovery from Data* 19.5 (2025), pp. 1–25.
- [5] Zhao, Tianyi\*, **Zhang, Liangliang\***, Y. Ma, and L. Cheng. “A Survey on Safe Multi-Modal Learning System”. In: *In Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (SIGKDD)* (2024). \* contributed equally to this work.
- [6] **Zhang, Liangliang**, M. Chen, X. Hu, and L. Deng. “Graph Convolutional Network and Contrastive Learning Small Nucleolar RNA (snoRNA) Disease Associations (GCLSDA): Predicting snoRNA–Disease Associations via Graph Convolutional Network and Contrastive Learning”. In: *International Journal of Molecular Sciences* 24.19 (2023), p. 14429.
- [7] D. Liu, X. Li, **Zhang, Liangliang**, X. Hu, J. Zhang, Z. Liu, and L. Deng. “Hgnnlda: predicting lncrna-drug sensitivity associations via a dual channel hypergraph neural network”. In: *IEEE/ACM transactions on computational biology and bioinformatics* 20.6 (2023), pp. 3547–3555.

## PREPRINTS AND SUBMISSIONS

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- [8] **Zhang, Liangliang**, N. Mihindukulasooriya, N. S. D’Souza, S. Shirai, S. Dash, Y. Ma, and H. Samulowitz. “From Factoid Questions to Data Product Requests: Benchmarking Data Product Discovery over Tables and Text”. In: *arXiv preprint arXiv:2510.21737* (2025).
- [9] H. Xu, **Zhang, Liangliang**, Y. Ma, S. Zhou, Z. Zheng, and B. Jiajun. “A Survey on Graph Condensation”. In: *arXiv preprint arXiv:2402.02000* (2024).

## RESEARCH EXPERIENCE

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**IBM Research**

Yorktown Heights, NY

*Research Intern*

*May 2026 – August 2026*

- Developed AgenticTT, an agentic framework for adaptive retrieval and reasoning over heterogeneous tables and text, where LLM agents dynamically select search, navigation, structured operations, and answer actions.

- Designed a typed evidence interface and reliability controls to support grounded multi-step reasoning across tables, rows, passages, and fused table–text evidence.
- Achieved 75.8 EM on OTT-QA, 68.2 EM on LakeQA-mini, and 70.6 EM on HybridQA, outperforming a matched fixed pipeline by 16.8 EM on OTT-QA and 43.8 EM on LakeQA-mini.
- Resulted in the paper “Adaptive Agentic Retrieval and Reasoning for Table–Text QA,” under review at AAAI 2027; arXiv preprint forthcoming.

## IBM Research

Yorktown Heights, NY

*Research Intern*

*May 2025 – August 2025*

- Developed a large-scale benchmark for data product discovery over tables and text, extending factoid retrieval to complex requests requiring coherent sets of structured and unstructured evidence.
- Constructed and validated more than 13K data product requests across Wikipedia and financial domains and evaluated BM25, dense, hybrid, and reranking retrieval methods.
- Analyzed retrieval performance across data product sizes and identified candidate generation and multi-source evidence coverage as major bottlenecks in heterogeneous retrieval.
- Resulted in the paper “From Factoid Questions to Data Product Requests: Benchmarking Data Product Discovery over Tables and Text” (arXiv:2510.21737).

## DAMI Lab, Rensselaer Polytechnic Institute

Troy, NY

*Graduate Research Assistant, advised by Prof. Yao Ma*

*Jan. 2025 – Present*

- Conduct research on **LLM retrieval and reasoning**, including KG-RAG, agentic retrieval, knowledge-graph reasoning, and reliable evaluation.
- Led a systematic analysis of **16 knowledge-graph QA benchmarks** and developed **KGQAGen-10K**, a benchmark with 10,787 LLM-generated and programmatically verified examples; published at **NeurIPS 2025**.
- Studied retrieval bottlenecks in knowledge-graph augmented reasoning by evaluating multiple graph retrieval systems and analyzing the gap between retrieved and gold evidence.
- Develop **planner–executor and agentic retrieval methods** for multi-step reasoning over knowledge graphs, tables, and text, with emphasis on adaptive evidence acquisition and reliable execution.

## Institute of Automation, Chinese Academy of Sciences

Beijing, China

*Research Intern*

*Mar. 2023 – Sep. 2023*

- Implemented and evaluated adversarial robustness methods for deep neural networks, including **CW**, **AutoAttack**, and **CopyCAT**, using PyTorch, Linux, and Docker.

## SELECTED HONORS & AWARDS

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- **Best Doctoral Consortium Presentation Award**, ACM WSDM 2026.
- **Student Travel Award**, ACM WSDM 2026.
- **Student Travel Award**, ACM SIGKDD 2024.
- **Academic Year Scholarship**, Central South University — awarded four times.

## TECHNICAL SKILLS

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**Programming:** Python, C++, SQL, Java

**Machine Learning / LLMs:** PyTorch, PyTorch Geometric, Hugging Face Transformers, vLLM, LangChain

**Retrieval / Data Systems:** FAISS, OpenSearch, Milvus, BM25, Dense Retrieval, Reranking

**Tools:** Linux, Docker, Git, LaTeX

## TEACHING EXPERIENCE

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### Rensselaer Polytechnic Institute

Troy, NY

*Teaching Assistant, CSCI 1100: Computer Science*

*Aug. 2024 – Dec. 2024*

- Led laboratory sessions and office hours and evaluated assignments and examinations for an introductory computer science course.